

Практикум

D5:archive

Twenty-eight questions, 97 marks. Every question the archive asks about a normally distributed variable, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

О практикуме

Нормальная половина
statistics.continuous_random_variables: 103 балла из 199 (доля 0.518) — всё, где у величины кривая с μ и σ , а не плотность формулой. И 4 балла statistics.probability (0.017 — 4 из 237): сумма трёх площадей ноября 2023 Q10(a). 107 баллов по корпусу, 97 рабочих: десять — ноябрьский дубль 2023 года. 80 баллов Paper 2, 10 Paper 3, 7 Paper 1; доля калькулятора 90 из 97 честная.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d5-normal

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D5 archive — the normal distribution, all of it

Twenty-eight questions, 97 marks.

Every question the archive asks about a normally distributed variable, from May 2021 to November 2025, in the order of the six techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `praciticum-d5-normal.ipynb`.

§	technique	questions	marks
1	The area	7	17
2	Areas that add up	4	11
3	A boundary from an area	6	18
4	One parameter	5	19
5	Two parameters	3	15
6	A condition inside	3	17

The checks are the same ones the practicum uses and they store nothing: each is handed the model and the facts of the question, finds the area under the

curve itself, and finds any unknown boundary, σ or μ itself. `Normal(mean, variance)` takes the **variance**, as IB writes $N(\mu, \sigma^2)$. Three significant figures, unless the question asks for two or four.

Parts of one question are split by technique. May 2025 TZ2 Q10 appears in §§ 1, 3 and 5; each part repeats what it needs. When a part uses a σ found in an earlier part, the check finds that σ from the question's own sentence, so an earlier slip does not follow you.

The November 2023 paper appears once, although the archive holds it twice as TZ1 and TZ2.

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit.py
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Normal, Mix, Freq, P(), SD

language('en')                 # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. The area

Seven questions, 17 marks. The model is known: shade the region, then the area.

1.1 — May 2021 TZ1 Paper 2 Q2(a)-(b), 3 marks

A company produces bags of sugar whose masses, in grams, can be modelled by a normal distribution with mean 1000 and standard deviation 3.5. A bag of sugar is rejected for sale if its mass is less than 995 grams.

(a) Find the probability that a bag selected at random is rejected.

(b) Estimate the number of bags which will be rejected from a random sample of 100 bags.

```

q1_1a = ...      # P(rejected)
q1_1b = ...      # the estimated number
rejected out of 100

sugar = Normal(1000, 3.5**2, 'X')

verify_chance('1.1(a)', q1_1a, P(sugar <
995))
verify_moment('1.1(b)', q1_1b,
Expect(Bin(100, P(sugar < 995))), count=True)

```

1.2 — May 2021 TZ2 Paper 2 Q10(b), 2 marks

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes, and 2% of the flight times are longer than 82 minutes. Part (a) finds σ — it is question 4.1 below.

(b) Find the probability that a randomly selected flight will have a flight time of more than 80 minutes.

```

q1_2 = ...          # P(T > 80)

sigma = symbols('sigma')
T = Normal(75, sigma**2, 'T')

verify_chance('1.2', q1_2, P(T > 80), given=
[Eq(P(T > 82), 0.02)], var=sigma)

```

1.3 — May 2022 TZ1 Paper 2 Q11(a), 2 marks

The weights, C grams, of the chocolate muffins made by a bakery are normally distributed with a mean of 62 g and standard deviation of 2.9 g.

(a) Find the probability that a randomly selected chocolate muffin weighs less than 61 g.

1.4 — November 2022 Paper 2 Q10(a), 2 marks

The time worked, T , in hours per week by employees of a large company is normally distributed with a mean of 42 and standard deviation 10.7.

(a) Find the probability that an employee selected at random works more than 40 hours per week.

```
q1_3 = ...      # P(C < 61)
q1_4 = ...      # P(T > 40)

verify_chance('1.3', q1_3, P(Normal(62,
2.9**2, 'C') < 61))
verify_chance('1.4', q1_4, P(Normal(42,
10.7**2, 'T') > 40))
```

1.5 — May 2023 TZ2 Paper 2 Q3(a), 2 marks

The weights, W grams, of bags of rice packaged in a factory can be modelled by a normal distribution with mean 204 grams and standard deviation 5 grams.

(a) A bag of rice is selected at random. Find the probability that it weighs more than 210 grams.

1.6 — May 2024 TZ1 Paper 2 Q3(a), 2 marks

The random variable X is normally distributed with mean 10 and standard deviation 2.

(a) Find the probability that X is more than 1.5 standard deviations above the mean.

```

q1_5 = ...          # P(W > 210)
q1_6 = ...          # P(X more than 1.5 standard
deviations above the mean)

X = Normal(10, 2**2, 'X')

verify_chance('1.5', q1_5, P(Normal(204,
5**2, 'W') > 210))
verify_chance('1.6', q1_6, P(X > 10 + 1.5*2))

```

1.7 — May 2025 TZ2 Paper 2 Q10(a), (c), 4 marks

At Adam's Apple Orchard the weights of apples, W , in grams, are normally distributed with a mean 175 grams and standard deviation 8 grams.

(a) Find the probability that a randomly chosen apple weighs less than 170 grams.

All orchards classify an apple as premium when its weight is between 170 and 185 grams.

(c) Find the percentage of apples that are classified as premium at Adam's Apple Orchard.

```
q1_7a = ...      # P(W < 170)
q1_7c = ...      # the percentage of premium
apples

W = Normal(175, 8**2, 'W')

verify_chance('1.7(a)', q1_7a, P(W < 170))
verify_chance('1.7(c)', q1_7c, P((W > 170) &
(W < 185)), percent=True)
```

§ 2. Areas that add up

Four questions, 11 marks. Symmetry, the total of one, and the rule for whole standard deviations — seven of these marks are on Paper 1.

2.1 — November 2023 TZ1 Paper 2 Q10(a), 2 marks

The height, H cm, of each wheat plant can be modelled by a normal distribution with mean m and standard deviation s . It is known that $P(H < 94.6) = 0.288$ and $P(H > 98.1) = 0.434$.

(a) Find the probability that the height of a randomly selected plant is between 94.6 cm and 98.1 cm.

2.2 — May 2023 TZ2 Paper 2 Q3(b), 2 marks

The weights, W grams, of bags of rice can be modelled by a normal distribution with mean 204 grams and standard deviation 5 grams. According to this

model, 80% of the bags of rice weigh between w grams and 210 grams.

(b) Find the probability that a randomly selected bag of rice weighs less than w grams.

```
q2_1 = ...      # P(94.6 < H < 98.1)
q2_2 = ...      # P(W < w)

m, s, w = symbols('m s w')
H = Normal(m, s**2, 'H')
wheat = [Eq(P(H < 94.6), 0.288), Eq(P(H >
98.1), 0.434)]
W = Normal(204, 5**2, 'W')

verify_chance('2.1', q2_1, P((H > 94.6) & (H
< 98.1)), given=wheat, var=[m, s])
verify_chance('2.2', q2_2, P(W < w), given=
[Eq(P((W > w) & (W < 210)), 0.8)], var=w)
```

2.3 — May 2024 TZ2 Paper 1 Q6(a), 1 mark

The weights of eating apples, in grams, can be modelled as a normal distribution with mean 100 g and standard deviation 20 g. You can assume that 95% of the weights are within two standard deviations of the mean.

(a) Find the percentage of eating apples that have a weight greater than 140 g.

2.4 — May 2025 TZ3 Paper 1 Q5, 6 marks

The random variables X and Y are normally distributed with $X \sim N(7, a^2)$ and $Y \sim N(19, a^2)$, where $a > 0$.

(a) Find b such that $P(X > b) = P(Y > 22)$.

(b) Write down the approximate value of $P(7 - a < X < 7 + a)$, correct to two significant figures.

(c) Given that $a = 3$, calculate the approximate value of $P(Y < 22)$, correct to two significant figures.

```

q2_3 = ...          # the percentage over 140 g
q2_4a = ...         # b
q2_4b = ...         #  $P(7 - a < X < 7 + a)$ , two
significant figures
q2_4c = ...         #  $P(Y < 22)$  when  $a = 3$ , two
significant figures

a, b = symbols('a b')
X = Normal(7, a**2, 'X')
Y = Normal(19, a**2, 'Y')

verify_chance('2.3', q2_3, P(Normal(100,
20**2, 'E', rule={2: 0.95})) > 140),
percent=True)
verify_letters('2.4(a)', q2_4a, b, [X, Y],
[Eq(P(X > b), P(Y > 22))], free=a)
verify_chance('2.4(b)', q2_4b, P((X > 7 - a)
& (X < 7 + a)), sf=2, free=a)
verify_chance('2.4(c)', q2_4c, P(Normal(19,
3**2, 'Y') < 22), sf=2)

```

§ 3. A boundary from an area

Six questions, 18 marks. Translate into the area to the left, then `invNorm`.

3.1 — May 2023 TZ2 Paper 2 Q3(c), 2 marks

The weights, W grams, of bags of rice can be modelled by a normal distribution with mean 204 grams and standard deviation 5 grams, and 80% of the bags weigh between w grams and 210 grams.

(c) Find the value of w .

3.2 — May 2025 TZ2 Paper 2 Q10(b), 2 marks

The weights of apples, W , in grams, are normally distributed with a mean 175 grams and standard deviation 8 grams.

(b) It is found that 20% of the apples weigh more than w grams. Find w , correct to four significant figures.

3.3 — May 2024 TZ1 Paper 2 Q3(b), 2 marks

The random variable X is normally distributed with mean 10 and standard deviation 2. The probability that X is more than k standard deviations above the mean is 0.1, where $k \in \mathbb{R}$.

(b) Find the value of k .

```
q3_1 = ...      # w for the rice
q3_2 = ...      # w for the apples, four
significant figures
q3_3 = ...      # k

w, k = symbols('w k')
rice = Normal(204, 5**2, 'W')
apples = Normal(175, 8**2, 'W')
X = Normal(10, 2**2, 'X')

verify_letters('3.1', q3_1, w, [rice],
[Eq(P((rice > w) & (rice < 210)), 0.8)])
verify_letters('3.2', q3_2, w, [apples],
[Eq(P(apples > w), 0.2)], sf=4)
verify_letters('3.3', q3_3, k, [X], [Eq(P(X >
10 + k*2), 0.1)])
```

3.4 — November 2025 TZ3 Paper 2 Q4(a), 3 marks

The heights of large sunflowers are normally distributed with mean m and

standard deviation s . It is known that 25 % of large sunflowers have a height of over 180 cm.

(a) Given that $m + As = 180$, where $A \in \mathbb{R}$, find the value of A .

```
q3_4 = ...      # A

m, s, A = symbols('m s A')
large = Normal(m, s**2, 'large')

verify_letters('3.4', q3_4, A, [large],
[Eq(P(large > 180), 0.25), Eq(m + A*s, 180)],
free=s)
```

3.5 — November 2025 TZ1 Paper 2 Q5, 5 marks

Consider the following grouped data set.

class	frequency
$15 < y \leq 20$	31
$20 < y \leq 25$	42
$25 < y \leq 30$	61
$30 < y \leq 35$	46
$35 < y \leq 40$	29

(a) Find an estimate for **(i)** the mean;
(ii) the standard deviation.

The data can be approximated by a normal distribution.

(b) Use this information and your mean and standard deviation found in part (a) to find an approximate value for the interquartile range correct to two significant figures.

```
q3_5m = ...      # the mean
q3_5s = ...      # the standard deviation
q3_5b = ...      # the interquartile range,
two significant figures

data = Freq({Interval.Lopen(15, 20): 31,
Interval.Lopen(20, 25): 42,
              Interval.Lopen(25, 30): 61,
Interval.Lopen(30, 35): 46,
              Interval.Lopen(35, 40): 29},
'y')
Y = Normal(Expect(data), Var(data), 'Y')
Q1, Q3, R = symbols('Q1 Q3 R')

verify_moment('3.5(a)(i)', q3_5m,
Expect(data))
verify_moment('3.5(a)(ii)', q3_5s, SD(data))
verify_letters('3.5(b)', q3_5b, R, [Y],
[Eq(P(Y < Q1), 0.25), Eq(P(Y < Q3), 0.75),
Eq(R, Q3 - Q1)], sf=2)
```

3.6 — November 2022 Paper 2

Q10(d), 4 marks

The time worked, T , in hours per week by employees of a large company is normally distributed with a mean of 42 and standard deviation 10.7.

It is known that $P(a \leq T \leq b) = 0.904$ and that $P(T > b) = 2P(T < a)$, where a and b are numbers of hours worked per week. An employee who works fewer than a hours per week is considered to be a part-time employee.

(d) Find the maximum time, in hours per week, that an employee can work and still be considered part-time.

```
q3_6 = ...          # a

a, b = symbols('a b')
T = Normal(42, 10.7**2, 'T')

verify_letters('3.6', q3_6, a, [T], [Eq(P((T
>= a) & (T <= b)), 0.904), Eq(P(T > b), 2*P(T
< a))])
```

§ 4. One parameter

Five questions, 19 marks. One letter in the model, one area: through $z = \frac{x-\mu}{\sigma}$.

4.1 — May 2021 TZ2 Paper 2 Q10(a), 3 marks

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes.

(a) Given that 2% of the flight times are longer than 82 minutes, find the value of σ .

4.2 — May 2023 TZ1 Paper 2 Q4, 5 marks

A company manufactures metal tubes for bicycle frames. The diameters of the tubes, D mm, are normally distributed with mean 32 and standard deviation s . The interquartile range of the diameters is 0.28.

Find the value of s .

```

q4_1 = ...      # sigma
q4_2 = ...      # s

sigma, s, Q1, Q3 = symbols('sigma s Q1 Q3')
T = Normal(75, sigma**2, 'T')
D = Normal(32, s**2, 'D')

verify_letters('4.1', q4_1, sigma, [T],
[Eq(P(T > 82), 0.02)])
verify_letters('4.2', q4_2, s, [D], [Eq(P(D <
Q1), 0.25), Eq(P(D < Q3), 0.75), Eq(Q3 - Q1,
0.28)])

```

4.3 — November 2023 TZ1 Paper 2 Q10(d), 3 marks

In another field, the farmer is growing the same variety of wheat, but is using a different fertilizer. The heights of these plants, F cm, are normally distributed with mean 98.6 and standard deviation d . The farmer finds the interquartile range to be 4.82 cm.

(d) Find the value of d .

4.4 — May 2025 TZ1 Paper 2 Q4(c), 3 marks

The heights, H cm, of adults in a group can be modelled by a normal distribution with mean 163 cm and standard deviation

s cm. It is found that 88% of the group have a height between 153 cm and 173 cm.

(c) Find the value of s .

```
q4_3 = ...      # d
q4_4 = ...      # s

d, s, Q1, Q3 = symbols('d s Q1 Q3')
F = Normal(98.6, d**2, 'F')
H = Normal(163, s**2, 'H')

verify_letters('4.3', q4_3, d, [F], [Eq(P(F < Q1), 0.25), Eq(P(F < Q3), 0.75), Eq(Q3 - Q1, 4.82)])
verify_letters('4.4', q4_4, s, [H], [Eq(P((H > 153) & (H < 173)), 0.88)])
```

4.5 — May 2022 TZ1 Paper 2 Q11(d), 5 marks

A bakery makes two types of muffins. The weights of the banana muffins are normally distributed with a mean of 68 g and standard deviation of 3.4 g. Each day 60% of the muffins made are chocolate.

The machine that makes the chocolate muffins is adjusted so that the mean weight of the chocolate muffins is 62 g but their standard deviation changes to σ

g. The machine that makes the banana muffins is not adjusted. The probability that the weight of a randomly selected muffin from these machines is less than 61 g is now 0.157.

(d) Find the value of σ .

```
q4_5 = ...          # sigma

sigma = symbols('sigma')
chocolate = Normal(62, sigma**2, 'C')
banana = Normal(68, 3.4**2, 'B')
muffin = Mix({chocolate: 0.6, banana: 0.4},
             'muffin')

verify_letters('4.5', q4_5, sigma,
              [chocolate], [Eq(P(muffin < 61), 0.157)])
```

§ 5. Two parameters

Three questions, 15 marks. Two areas, two z -values, two linear equations.

5.1 — November 2023 TZ1 Paper 2 Q10(b), 5 marks

The height, H cm, of each wheat plant can be modelled by a normal distribution with mean m and standard deviation s . It is known that $P(H < 94.6) = 0.288$ and $P(H > 98.1) = 0.434$.

(b) Find the value of m and the value of s .

5.2 — May 2025 TZ2 Paper 2 Q10(f), 6 marks

An apple is classified as premium when its weight is between 170 and 185 grams. At a neighbouring orchard the weights of apples, M , in grams, are normally distributed with mean m and standard deviation s . It is known that:

- 82% of their apples are classified as premium;

- the percentage of apples that weigh less than 170 grams is twice the percentage of apples that weigh more than 185 grams.

(f) Find the value of m .

```
q5_1 = [...]      # [m, s]
q5_2 = ...        # m

m, s = symbols('m s')
H = Normal(m, s**2, 'H')
M = Normal(m, s**2, 'M')

verify_letters('5.1', q5_1, [m, s], [H],
[Eq(P(H < 94.6), 0.288), Eq(P(H > 98.1),
0.434)])
verify_letters('5.2', q5_2, m, [M], [Eq(P((M
> 170) & (M < 185)), 0.82), Eq(P(M < 170),
2*P(M > 185))])
```

5.3 — November 2025 TZ3 Paper 2 Q4(b), 4 marks

The heights of large sunflowers are normally distributed with mean m and standard deviation s , and 25% of large sunflowers have a height of over 180 cm.

The giant sunflowers have a mean height that is 35 cm more than that of the large sunflowers and a standard deviation that

is double that of the large sunflowers. Their heights are also normally distributed, and 98% of giant sunflowers have a height greater than 180 cm.

(b) Determine m and s .

```
q5_3 = [...]      # [m, s]

m, s = symbols('m s')
large = Normal(m, s**2, 'large')
giant = Normal(m + 35, (2*s)**2, 'giant')

verify_letters('5.3', q5_3, [m, s], [large,
giant], [Eq(P(large > 180), 0.25), Eq(P(giant
> 180), 0.98)])
```

§ 6. A condition inside

Three questions, 17 marks. A ratio of areas, and an event about another quantity turned into an interval of Z .

6.1 — May 2021 TZ1 Paper 2 Q2(c), 3 marks

The masses of bags of sugar, in grams, can be modelled by a normal distribution with mean 1000 and standard deviation 3.5. A bag is rejected for sale if its mass is less than 995 grams.

(c) Given that a bag is not rejected, find the probability that it has a mass greater than 1005 grams.

6.2 — May 2021 TZ2 Paper 2 Q10(c), 4 marks

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes, and 2% of the flight times are longer than 82 minutes.

(c) Given that a flight between the two cities takes longer than 80 minutes, find the probability that it takes less than 82 minutes.

```
q6_1 = ...          # P(X > 1005 | not rejected)
q6_2 = ...          # P(T < 82 | T > 80)

sugar = Normal(1000, 3.5**2, 'X')
sigma = symbols('sigma')
T = Normal(75, sigma**2, 'T')

verify_chance('6.1', q6_1, P(sugar > 1005,
given=sugar >= 995))
verify_chance('6.2', q6_2, P(T < 82, given=T
> 80), given=[Eq(P(T > 82), 0.02)],
var=sigma)
```

6.3 — May 2024 TZ2 Paper 3 Q2(g)-(h), 10 marks

Consider a randomly generated quadratic function, $f(x) = x^2 + 2Zx + 1$, where the continuous random variable $Z \sim N(0, 1)$.

(g) Find the probability that the graph of f has two x -intercepts.

The continuous random variables, X_1 and X_2 , represent the x -intercepts of the

graph of f where $X_1 = -Z - \sqrt{Z^2 - 1}$ and $X_2 = -Z + \sqrt{Z^2 - 1}$.

(h) Given that the graph of f has two x -intercepts, X_1 and X_2 , find the probability that both X_1 and X_2 are greater than 0.5.

```
q6_3g = ...      # P(two x-intercepts)
q6_3h = ...      # P(both > 0.5 | two x-
intercepts)

Z = Normal(0, 1, 'Z')
X1 = Z.map(lambda z: -z - sqrt(z**2 - 1),
'X1')
X2 = Z.map(lambda z: -z + sqrt(z**2 - 1),
'X2')

verify_chance('6.3(g)', q6_3g, P(X1 < X2))
verify_chance('6.3(h)', q6_3h, P((X1 > 0.5) &
(X2 > 0.5), given=X1 < X2))
```



Solutions

1.1 $P(X < 995) = 0.0765637 \dots = \boxed{0.0766}$; $100 \times 0.0766 = \boxed{7.66}$, and the markscheme accepts 8 bags.

1.2 $\sigma = \frac{7}{2.05374 \dots} = 3.40840 \dots$, and $P(T > 80) = \boxed{0.0712}$.

1.3 $P(C < 61) = 0.365111 \dots = \boxed{0.365}$.

1.4 $P(T > 40) = 0.574136 \dots = \boxed{0.574}$.

1.5 $P(W > 210) = 0.115069 \dots = \boxed{0.115}$.

1.6 $10 + 1.5 \times 2 = 13$: $P(X > 13) = 0.0668072 \dots = \boxed{0.0668}$.

1.7 $P(W < 170) = \boxed{0.266}$; $P(170 < W < 185) = 0.628364 \dots$, $\boxed{62.8 \%}$.

2.1 $1 - 0.288 - 0.434 = \boxed{0.278}$.

2.2 $1 - 0.8 - 0.115069 \dots = \boxed{0.0849}$.

2.3 $140 = \mu + 2\sigma$; outside $\mu \pm 2\sigma$ is 5 %, half of it above: $\boxed{2.5 \%}$.

2.4 (a) 22 is 3 above 19, so $b = 7 + 3 = \boxed{10}$.
(b) $\boxed{0.68}$. (c) $22 = 19 + 3$ is one standard deviation up: $1 - \frac{0.32}{2} = \boxed{0.84}$.

3.1 $w = \text{invNorm}(0.0849302\dots, 204, 5) = 197.136\dots = \boxed{197}$.

3.2 Area to the left 0.8: $w = 181.732\dots = \boxed{181.7}$.

3.3 $P(Z > k) = 0.1$: $k = \text{invNorm}(0.9) = \boxed{1.28}$.

3.4 $P(Z < A) = 0.75$: $A = \boxed{0.674}$.

3.5 Mid-interval values, $\sum f = 209$: mean $\boxed{27.5}$, $\sigma = 6.26374\dots = \boxed{6.26}$. Quartiles of $N(27.5, 6.26374^2)$: 23.2751... and 31.7248..., IQR $8.44965\dots = \boxed{8.4}$.

3.6 With $P(T < a) = x$: $x + 0.904 + 2x = 1$, $x = 0.032$, and $a = \text{invNorm}(0.032, 42, 10.7) = 22.1816\dots = \boxed{22.2}$ (the markscheme also accepts 22.1).

4.1 $\frac{82-75}{\sigma} = 2.05374\dots$: $\sigma = 3.40840\dots = \boxed{3.41}$.

$$\mathbf{4.2} \quad Q_3 = 32.14: \quad \frac{0.14}{s} = 0.674489 \dots, \quad s = 0.207564 \dots = \boxed{0.208}.$$

$$\mathbf{4.3} \quad Q_3 = 101.01: \quad \frac{2.41}{d} = 0.674489 \dots, \quad d = 3.57307 \dots = \boxed{3.57}.$$

$$\mathbf{4.4} \quad 6\% \text{ above } 173: \quad \frac{10}{s} = 1.55477 \dots, \quad s = 6.43181 \dots = \boxed{6.43}.$$

$$\mathbf{4.5} \quad 0.6 P(C < 61) + 0.4 \times 0.0197555 \dots = 0.157 \text{ gives } P(C < 61) = 0.248496 \dots; \quad z = -0.679229 \dots, \quad \frac{61-62}{\sigma} = z, \quad \sigma = 1.47225 \dots = \boxed{1.47}.$$

$$\mathbf{5.1} \quad m - 0.559236 \dots s = 94.6 \quad \text{and} \quad m + 0.166199 \dots s = 98.1: \quad \boxed{m = 97.3, s = 4.82}.$$

$$\mathbf{5.2} \quad \text{Outside the premium range is } 18\%, \text{ split two to one: with } P(M > 185) = x, \quad 2x + 0.82 + x = 1, \text{ so } P(M < 170) = 0.12 \text{ and } P(M > 185) = 0.06. \quad z = -1.17498 \dots \text{ and } 1.55477 \dots: \quad m - 1.17498 \dots s = 170, \quad m + 1.55477 \dots s = 185, \text{ so } s = 5.49498 \dots \text{ and } m = 176.456 \dots = \boxed{176}.$$

$$\mathbf{5.3} \quad m + 0.674489 \dots s = 180 \text{ and } \frac{180 - (m + 35)}{2s} = -2.05374 \dots: \quad s = 7.31913 \dots, \quad m = 175.063 \dots,$$

$$\boxed{m = 175, s = 7.32}.$$

$$\mathbf{6.1} \quad \frac{P(X > 1005)}{P(X \geq 995)} = \frac{0.0765637\dots}{0.923436\dots} = \boxed{0.0829}.$$

$$\mathbf{6.2} \quad \frac{P(80 < T < 82)}{P(T > 80)} = \frac{0.0511929\dots}{0.0711929\dots} = \boxed{0.719}.$$

$$\mathbf{6.3} \quad (\text{g}) \quad 4Z^2 - 4 > 0: P(|Z| > 1) = \boxed{0.317}. \quad (\text{h})$$

$X_1 > 0.5$ needs $Z < -1$ and $Z > -1.25$:

$$\frac{0.0530054\dots}{0.317310\dots} = \boxed{0.167}.$$